

```
In [2]: import pandas as pd
        from sklearn.model_selection import train_test_split
        from sklearn.tree import DecisionTreeClassifier, export_text
        from sklearn.metrics import accuracy_score, confusion_matrix
```

```
In [3]: df=pd.read_csv('IRIS.csv')
        df
```

```
Out[3]:
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Iris-virginica
146	6.3	2.5	5.0	1.9	Iris-virginica
147	6.5	3.0	5.2	2.0	Iris-virginica
148	6.2	3.4	5.4	2.3	Iris-virginica
149	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 5 columns

```
In [4]: df.nunique()
```

```
Out[4]: sepal_length    35
        sepal_width    23
        petal_length   43
        petal_width    22
        species         3
        dtype: int64
```

```
In [6]: X = df[['sepal_length', 'sepal_width', 'petal_length', 'petal_width']
        Y = df['species']
```

```
In [7]: X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.3)
```

```
In [8]: dtree = DecisionTreeClassifier(criterion="gini", max_depth=3, random_state=42)
        dtree.fit(X_train, Y_train)
```

```
Out[8]: ▼ DecisionTreeClassifier ⓘ ?
        ► Parameters
```

```
In [9]: Y_pred = dtree.predict(X_test)
```

```
In [10]: print("Accuracy:", accuracy_score(Y_test, Y_pred))
print("\nConfusion Matrix:\n", confusion_matrix(Y_test, Y_pred))
```

Accuracy: 1.0

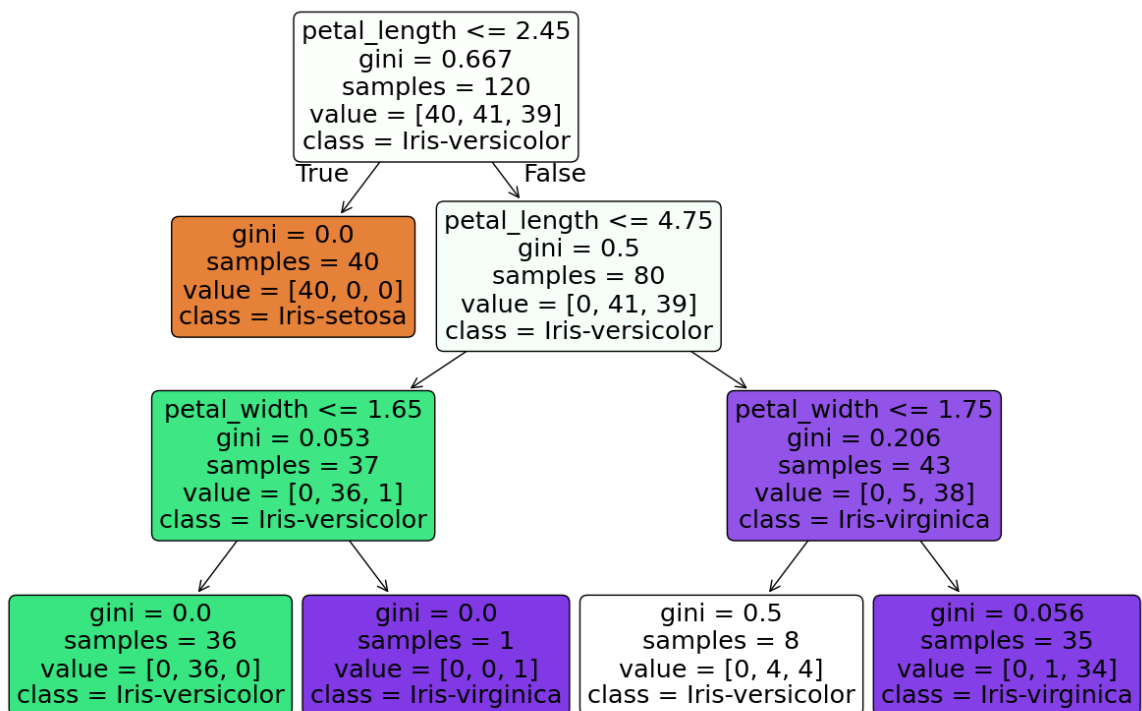
Confusion Matrix:

```
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

```
In [11]: import matplotlib.pyplot as plt
from sklearn.tree import plot_tree

plt.figure(figsize=(15,10))

plot_tree(dtree,
          feature_names=X.columns,
          class_names=dtree.classes_,
          filled=True,
          rounded=True
        )
plt.show()
```



```
In [ ]:
```